

Every Natural Strengthening Examined Is False: Type Mismatch in Lossless Reformulations of the Lonely Runner Conjecture

A sharp negative result with a mechanism: four settings across two reformulation worlds—the finite covering reformulation, the transfer-matrix cascade, character rigidity, and the Lonely Runner zonotope—in which every strengthening that would open a new toolkit and was examined is false, with exact rational witnesses ($16/47 > 1/3$ at the harmonic instance—measured exactly, from $n = 5$ on). Revision (v4): the harmonic instance certified— $p(1,2,3,4,5) = 16/47$ exactly, by a balance-family witness whose three tied dips obey a weighted identity, and a branch-and-bounds tree emptied by two new certificates (a sliding- σ interval test and a triple-identity certificate); the two $n = 6$ boundary rows resolved as further strict refutations ($17/47$ and $455/1269 > 5/14$). With the program's bounded positive theorem and a certified deepest-hole archive.

Anonymous

Reformulation Program, (companion to the pair-sum monograph)

A negative-results paper: no solution, no new cases of the conjecture, the surviving equivalent problems as hard as they ever were.